# OrbitalWatch: AI-Assisted Indications & Warning for Sub-Threshold Threats to Space Systems

*Research concept note · 2026-11-05*


## Abstract

OrbitalWatch is the technical half of a two-part research program; its companion is a governance study of a satellite ground-segment / PNT timing compromise and its cross-border consequences. The two are unified by one question, the integrity of signals derived from space systems, approached from opposite ends.

OrbitalWatch is an AI-assisted indications-and-warning (I&W) tool that detects sub-threshold, gray-zone hostile satellite behavior (rendezvous and proximity operations, shadowing, co-orbital approaches) using only public orbital-tracking (TLE) data. It scores anomalies, separates benign orbital change from powered maneuvers with a physics-grounded and explainable classifier, maps detections to MITRE ATT&CK for Space, and wraps the result in a governance structure that turns a detection into an auditable warning. The research contribution is that governance wrapper, not the detector alone. Methods are written; results are pending analysis runs and are not reported here.


## 1. Problem & Gap

Hostile activity in orbit increasingly takes a sub-threshold, gray-zone form: not kinetic strikes but rendezvous and proximity operations, shadowing, and co-orbital approaches that intimidate, surveil, or pre-position without clearly crossing a line. This behavior is hard to attribute and easy to deny, which is much of its value to an adversary.

Defenders who are not inside a classified space-domain-awareness enterprise have little accessible, transparent indications-and-warning (I&W) to draw on. Classified sensing exists, but it is not shareable, auditable, or usable by the researchers, operators, and allied partners who also need warning. The gap is an open, explainable pipeline that turns public orbital data into governed, defensible warnings.

That this behavior is real and current is shown by the Cosmos 2612 episode of autumn 2026, in which a Russian military satellite conducted repeated close-proximity maneuvers toward a U.S.-registered commercial radar satellite: a documented, dated event against which an open detection pipeline can be tested.

## 2. Research Questions

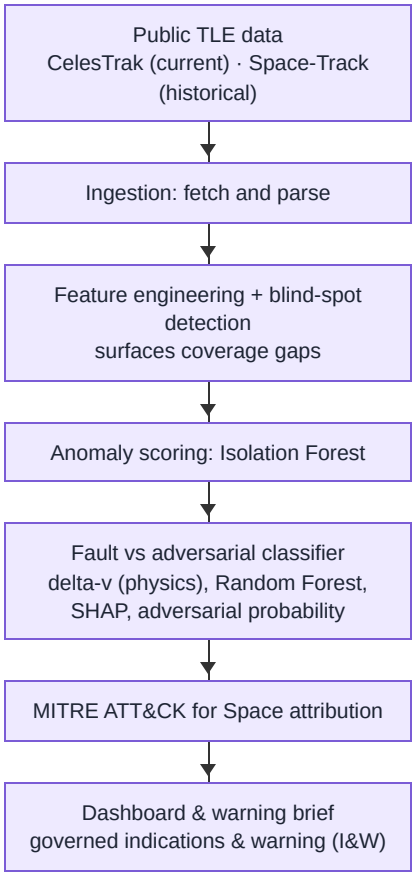
1. **Detection.** Can adversarial maneuvers (RPO, co-orbital approaches, plane-matching) be detected from public TLE data alone, despite its latency and gaps?
2. **Discrimination.** Can benign orbital change (decay, drag, routine station-keeping) be separated from powered, potentially hostile maneuvers using physics-grounded, explainable features rather than a black-box score?

3. **Attribution vocabulary.** Can detections be mapped to a shared threat vocabulary (MITRE ATT&CK for Space) and turned into a governed, auditable warning rather than a raw alert?
4. **Reliability.** How does the pipeline perform against a documented episode (Cosmos 2612 / ICEYE-X36) whose timeline is independently recorded in the public record?

## 3. System & Approach

OrbitalWatch is a pipeline that turns open orbital data into a governed warning. Public two-line element (TLE) data is ingested from CelesTrak (current) and Space-Track (historical), parsed, and turned into engineered orbital features. A blind-spot layer flags where tracking coverage is sparse, so gaps in the data are surfaced rather than hidden. An Isolation Forest scores each object for anomaly; flagged objects pass to a physics-grounded classifier that separates natural orbital change from powered maneuvers and returns an adversarial-probability score with an explanation. Detections are mapped to MITRE ATT&CK for Space so they speak a shared threat vocabulary, and a blast-radius estimate (extending the BlastMap approach) indicates ground-segment exposure. The output is a Streamlit dashboard and an auditable warning brief.

The pipeline runs top to bottom:




Each stage is explainable, and every input is public, so a warning can be shown, checked, and defended.

## 4. Detection Methods

Detection runs in stages, each chosen to stay explainable and to run on public data.

- **Feature engineering.** Each object's TLE history is turned into orbital features (changes in mean motion, inclination, eccentricity, and derived altitude) that capture how an orbit is evolving over time.
- **Anomaly scoring.** An Isolation Forest scores objects for how unusual their recent behavior is, without needing labeled attack data. This is a wide net, not a verdict.
- **Fault vs. adversarial classification.** Flagged objects pass to a physics-grounded classifier that estimates the delta-v (velocity change) implied by an orbit shift using orbital mechanics, separating natural change (decay, atmospheric drag, routine station-keeping) from powered maneuvers. A Random Forest with SHAP explanations returns an adversarial-probability score alongside the features that drove it, so an analyst sees *why* something was flagged.
- **Proximity analysis.** For co-orbital cases, SGP4 propagation (via the `sgp4` library) reconstructs relative motion between two objects to measure approach distance and phase.
- **Attribution vocabulary.** Detections are mapped to MITRE ATT&CK for Space so they are described in a shared, recognizable threat language.

Every step uses public inputs and exposes its reasoning; nothing depends on classified feeds.

## 5. Validation: the Cosmos 2612 Case Study

The pipeline is tested against a dated, independently documented episode rather than only synthetic data. Over late September and October 2026, the Russian military satellite Cosmos 2612 (NORAD 68762), one of six (Cosmos 2609–2614) launched in April 2026, conducted repeated close-proximity maneuvers toward ICEYE-X36 (NORAD 59103), a U.S.-registered synthetic-aperture-radar satellite widely reported to support Ukraine, matching its orbital plane and closing to within a few hundred meters.

A case-study module reconstructs the episode from public data: it fetches historical TLEs from Space-Track, propagates both objects with SGP4, detects the maneuvers, classifies the phases of relative motion, and scores the reconstructed timeline against the public record. A companion forward-watch tracks the sister satellites for similar behavior.

**Results are pending analysis runs and are not reported here.** The value of the episode is that its timeline is externally recorded, so the pipeline's output can be checked against something real rather than graded on its own synthetic test. No performance numbers should appear until the runs are complete.

## 6. Governance Framing

The research contribution is the governance structure around the detector, not the detector itself. A raw anomaly score is not a warning; it becomes one only when it is defined, bounded, and auditable. OrbitalWatch supplies that wrapper:

- **Defined indicators.** An explicit catalog of what counts as an indication, and what does not, so detections are judged against stated criteria, not analyst intuition.

- **An indications-and-warning discipline.** A repeatable path from signal to assessed warning, carrying confidence and uncertainty through rather than dropping them.
- **An auditable warning brief.** A standard format that records what was seen, how confident the system is, and what it cannot say, so a reader can check the warning, not just receive it.
- **Attribution guardrails.** Explicit limits on what the tool may claim, keeping "possible hostile maneuver" from hardening into "attributed attack."

This framing connects to responsible-AI governance for security tools and to the NIST AI Risk Management Framework, including its critical-infrastructure direction. The scholarly question is how to govern an AI detector well enough to issue warnings responsibly, not merely whether the detector fires.

## 7. Expected Contributions

1. An **open, explainable I&W pipeline** that derives warnings about hostile satellite behavior from public TLE data.
2. A **physics-grounded fault-vs-adversarial classifier** that separates natural orbital change from powered maneuvers and shows its reasoning (delta-v features, SHAP).
3. A **mapping to MITRE ATT&CK for Space**, giving detections a shared threat vocabulary.
4. A **governance wrapper** (defined indicators, an auditable warning brief, and uncertainty discipline) offered as the central research contribution.
5. **Validation against a documented episode** (Cosmos 2612 / ICEYE-X36), testing the pipeline against the public record.

## 8. Scope, Assumptions & Caveats

- **Public data only.** OrbitalWatch uses unclassified TLE data and open incident reporting; it draws on no classified space-domain-awareness feeds. That is a deliberate design choice (it is what makes the warnings shareable and auditable) and a limit on what it can see.
- **TLE limitations, stated plainly.** Public orbital data has latency, gaps, and limited accuracy; short or subtle maneuvers can be missed, and timing is approximate. The analysis accounts for this rather than assuming clean data.
- **Probabilistic, not proof.** A high adversarial-probability score is an indication, not evidence of intent. Attribution is handled cautiously and bounded by the governance guardrails above.
- **Decision support, not decision-maker.** The tool informs analysts; it does not make attribution or targeting decisions.
- **Results pending.** The methods are written; quantitative results await the analysis runs, and none are claimed until then.

## 9. Relationship to the Governance Companion

OrbitalWatch is the technical half of a two-part research program. Its companion is a consequence-and-governance study of a satellite ground-segment / PNT timing compromise and its cross-border impact

(Florida and the Bahamas). The two are unified by one question, the integrity of signals derived from space systems, approached from opposite ends: OrbitalWatch asks whether a satellite's observed *behavior* can be trusted; the companion asks what happens when infrastructure can no longer trust the *timing signal* a space system provides.

Together they model a **coordinated campaign**, a capable adversary acting in both segments at once. OrbitalWatch supplies indications and warning for the space segment, the orbital behavior it is built to flag, while the companion follows a parallel ground-segment action to its cross-border consequence and the governance gap that follows.

**One honest seam.** OrbitalWatch watches the space segment; it would *not* detect the ground-segment cyber action in the companion's scenario. The pairing is therefore a claim about adversary reach, and about coverage across two segments usually studied separately, not a claim that one tool sees the whole campaign.

## 10. Status & Development Path

**Status.** The tool exists (ingestion, anomaly detection, the fault-vs-adversarial classifier, MITRE ATT&CK for Space mapping, and a Streamlit dashboard), and the paper `orbitalwatch.tex` has its methods written. The results sections are pending analysis runs.

**Publication.** The goal is to publish both the tool and the research: an open GitHub repository, a Zenodo DOI, a preprint, and a peer-reviewed venue.

**Next steps.**

1. Run the analyses and fill the results sections (paper and the Cosmos 2612 case study).
2. Finalize the case study against the public timeline.
3. Line up an independent reviewer for publication.
4. Prepare the open-source release package.